
**Division 10-50.70:
Outdoor Lighting Standards**

Sections:

10-50.70.010	Purpose
10-50.70.020	Compliance with the Zoning Code and Permit Requirements
10-50.70.030	Applicability
10-50.70.040	Establishment of Lighting Zones
10-50.70.050	General Requirements – All Lighting Zones
10-50.70.060	Special Uses
10-50.70.070	Prohibited Outdoor Lighting
10-50.70.080	Exceptions
10-50.70.090	Nonconforming Outdoor Lighting
10-50.70.100	Violations and Enforcement

10-50.70.010 Purpose

The City was recognized as the world's first International Dark Sky City on October 24, 2001, for its pioneering development and implementation of lighting codes that balance the need to preserve Flagstaff's dark sky resource with the need for safe outdoor lighting practices. This division is to help assure that dark skies remain a resource to be enjoyed by the Flagstaff community and its visitors, minimize light pollution and Light Trespass across property lines, maintain an environment that the City's astronomical industry depends upon, reduce energy consumption, minimize the potential disruption to nocturnal ecosystems, and to allow safe and efficient outdoor lighting.

(Ord. 2022-27, Amended, 11/15/2022 (Res. 2022-51))

10-50.70.020 Compliance with the Zoning Code and Permit Requirements

A. An outdoor lighting permit shall be obtained prior to the installation, modification, or replacement of any Outdoor Light Fixture subject to Section [10-20.40.100](#).

1. Any modification of an existing Outdoor Light Fixture shall comply with this division.
2. All outdoor lighting shall be installed in compliance with the provisions of this Division [10-50.70](#); Title 4, Building Regulations; and Title 5, Fire Code, of the City Code.

(Ord. 2022-27, Amended, 11/15/2022 (Res. 2022-51))

10-50.70.030 Applicability

A. **New Development and Parking Areas.** The provisions of this division shall apply to any new development of a vacant lot or parcel and any installation of outdoor lighting to support that development, including parking areas and areas within an unenclosed parking garage.

B. **Major Additions, Modifications, Replacements, and Change of Use.** The provisions of this division shall apply to all major additions, modifications, replacements, or change of use. The entire property shall comply with the requirements of this division when any of the following occur:

1. "Major" shall be defined as an increase of 25 percent or more in terms of additional dwelling units,

gross floor area, seating capacity, or parking spaces, either with a single addition or with cumulative additions after the effective date of this provision; or

2. Single or cumulative modifications or replacements of legally installed Outdoor Light Fixtures after August 5, 1999, constituting 25 percent or more of the Lumens that would be permitted under this division for the property, regardless of the Total Outdoor Light Output currently existing on a site.

C. **Minor Additions, Modifications, Replacements, and Change of Use.** The provisions of this division shall not apply to minor (less than 25 percent as referenced above) additions, modifications, replacements, or change of use after August 5, 1999, except for the following:

1. All new or modified Outdoor Light Fixtures shall comply with the requirements of this division.
2. The Total Outdoor Light Output shall not exceed the amount allowed by this division, or the total legal nonconforming light output, whichever is greater.

D. All Outdoor Light Fixtures on property or buildings that are owned and operated by the City of Flagstaff shall be fully shielded.

(Ord. 2022-27, Amended, 11/15/2022 (Res. 2022-51); Ord. 2016-07, Amended, 2/16/2016 (Res. 2016-02))

10-50.70.040 Establishment of Lighting Zones

A. **Establishment of Lighting Zones.** Two lighting zones are established, with varying development standards specific to their location within the City. Lighting zones are shown in Section [10-90.40.020](#), Lighting Zone Map.

B. **Lighting Zone Boundaries.** The boundaries of the lighting zones are illustrated in Section [10-90.40.020](#), Lighting Zone Map.

1. Lighting Zone 1 includes all areas within the City of Flagstaff that are within two and one-half miles of the Kaj Strand telescope at the U.S. Naval Observatory.
2. Lighting Zone 2 includes all areas more than two and one-half miles of the Kaj Strand telescope at the U.S. Naval Observatory.

C. **Split Parcels.** A parcel located in more than one of the described zones shall be considered to be only in the more restrictive lighting zone.

(Ord. 2022-27, Amended, 11/15/2022 (Res. 2022-51))

10-50.70.050 General Requirements – All Lighting Zones

A. **Lighting Classes.** The Total Outdoor Light Output for a lot or parcel is cumulative of the light emitted by all lighting classes.

1. Class 1 Lighting is lighting used for applications where color rendition is required to preserve the effectiveness of an activity. All Class 1 Lighting shall have a correlated color temperature (CCT) of 2,700 Kelvin (K), or less. Recognized Class 1 Lighting applications include only the following examples:

- a. Outdoor sales areas, including service station canopies;

- b. Primary customer building entry/exit areas of commercial and industrial uses (does not include service or emergency entry/exits);
 - c. Outdoor eating areas at restaurants;
 - d. Outdoor assembly or repair areas where assembly or repair work occurs at night as part of the normal business operations;
 - e. Outdoor recreational field/track/arena areas; and
 - f. External and internal lighting for signs.
2. Class 2 Lighting is lighting used for applications where general illumination for safety or security is the primary concern. Narrow-Spectrum Amber LEDs are required in all Class 2 Lighting applications. Examples of Class 2 Lighting applications include the following:
- a. Pedestrian walkways, driveways, and roadways;
 - b. Parking lots;
 - c. Equipment yards; and
 - d. Outdoor security.
3. Class 3 Lighting is outdoor lighting used for decorative purposes. All light sources may be used for Class 3 Lighting including technologies that allow for the color of the light to change so long as the colors change less than every two minutes. Examples of Class 3 Lighting applications include the following:
- a. Architectural illumination;
 - b. Flag and monument lighting; and
 - c. Landscape lighting and the illumination of trees, shrubs, and other vegetation.

B. Total Outdoor Light Output.

1. The Total Outdoor Light Output shall not exceed the amounts allowed in Table 10-50.70.050.B.

Table 10-50.70.050.B			
Maximum Total Outdoor Light Output Standards			
Land Use		Lighting Zone 1	Lighting Zone 2
Commercial, Industrial, and Multiple-Family Development (Lumens per Net Acre)			
	Combined maximum for Fully and Partially Shielded Fixtures, and fixtures mounted to a building or canopy		
		17,500	35,000
	Partially Shielded Fixtures maximum	0	3,850
	Fixtures not mounted to a building or canopy ¹ maximum	4,000	35,000

Single-Family and Duplex Developments, Including Accessory Structures (Lumens per Lot, or Parcel)			
	Combined maximum for Fully and Partially Shielded Fixtures, and fixtures not mounted to a building or canopy, excluding motion sensing Outdoor Light Fixtures ²		
		1,500	5,000
	Partially Shielded Fixtures maximum	0	2,000
	Motion Sensing Outdoor Light Fixtures ² (fully shielded) maximum	0	2,000
	Fixtures that are not mounted to a building or canopy ¹ maximum	0	1,500
End Note			
1. Fixtures that are not directly mounted to the side of a building or to the underside of, or within, a canopy or overhang. 2. Motion sensing Outdoor Light Fixtures shall have an automated timer set to turn off the fixture at a time no greater than five minutes after the light has been turned on.			

C. **Shielding.** The standards provided in Table 10-50.70.050.C, Shielding Standards, shall apply:

Table 10-50.70.050.C		
Shielding Standards		
Land Use and Lighting Class	Lighting Zone 1	Lighting Zone 2
Commercial, Industrial, and Multiple-Family Development		
Class 1 Lighting All Outdoor Light Fixtures	FS	FS
Class 2 Lighting All Outdoor Light Fixtures	FS	FS
Class 3 Lighting		
All Outdoor Light Fixtures with a Lumen output below 1,750 ¹ Lumens	FS	A ¹
All Outdoor Light Fixtures with a Lumen output equal to and above 1,750 ¹ Lumens	X	X
Single-Family and Duplex Residential Dwelling Units		
Class 1 – 3 Lighting		
All Outdoor Light Fixtures with a Lumen output below 700 ¹ Lumens	FS	A ¹
All Outdoor Light Fixtures with a Lumen output equal to and above 700 ¹ Lumens	X	X

	End Notes
1.	For purposes of determining the Total Outdoor Light Output from an Outdoor Light Fixture, lighting assemblies which include multiple unshielded or Partially Shielded Fixtures or lamps on a single pole or assembled as a single unit shall be considered as one fixture. Refer to the Table 10-50.70.050.A .
	Key
FS = Allowed; Only Fully Shielded Fixtures A = Allowed; Fully Shielded Fixtures preferred, and Partially Shielded Fixtures are allowed subject to the Lumen amounts listed in Table 10-50.70.050.B X = Outdoor light fixture is prohibited	

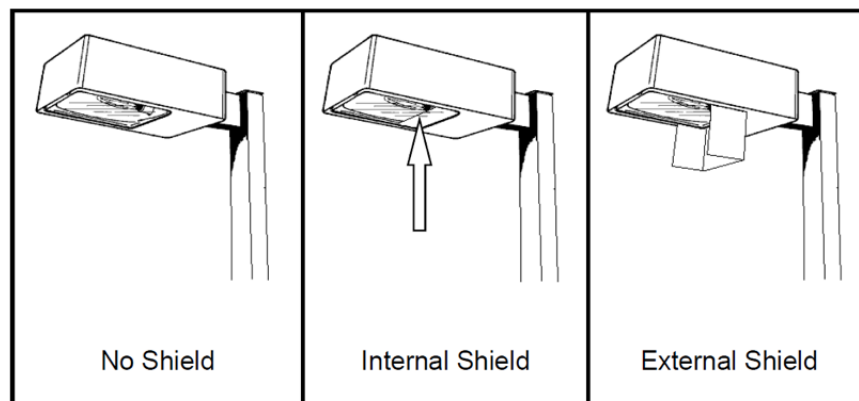
D. **Effective Shielding.** All Outdoor Light Fixtures that are required to be fully shielded shall be installed in such a manner that the shielding satisfies the definition of a Fully Shielded Light Fixture.

E. **Light Trespass Standard.**

1. All Outdoor Light Fixtures, including motion sensing lighting, shall be located, aimed, and shielded so that the direct illumination from the fixture shall be confined to the property boundaries of the source.
2. Any privately or publicly owned Outdoor Light Fixture with an output over 10,000 Lumens located within 50 feet of any residential (including Multiple-Family Development) property or public right-of-way shall utilize an internal or external shield, with the Outdoor Light Fixture and shield oriented to prevent direct illumination over the adjacent property or right-of-way line. If an external shield is used, its surface must be painted black to minimize reflections (Figure 10-50.70.050.C).

Figure 10-50.70.050.E

Shielding Configurations



F. **Motion Sensing Outdoor Light Fixtures.** Motion sensing Outdoor Light Fixtures shall be fully shielded.

G. **Time Limits for Outdoor Lighting.** All outdoor Class 1 and Class 3 Lighting, and outdoor Class 2 Lighting located more than 50 feet from any structure, outdoor product display, or storage area shall be turned off no later than 30 minutes after the business closes and are to remain off until the business reopens. Decorative holiday lights are exempt in accordance with Section 10-50.70.080.D, and outdoor sports facilities are subject to the time limits in Table 10-50.70.060.C.

H. **Sign Illumination.** Standards for external and internal sign illumination are provided in Division [10-](#)

[50.100](#), Sign Standards. Lighting used for the external illumination of signs is included toward the Total Outdoor Light Output standards of subsection B. of this section.

I. **Neon Building Lighting.** Neon or simulated LED neon building lighting is included in the Total Outdoor Light Output calculations for the site. Lumens for neon or simulated LED neon lighting are calculated on a per foot basis, rather than per “fixture.” Unshielded neon or simulated LED neon lighting is not allowed except for signage.

J. **Multi-Class Lighting.** Multi-class lighting must either conform to the lamp-type and shielding requirements of the strictest included class as shown in Table 10-50.70.050.B, Maximum Total Outdoor Light Output Standards, and Table 10-50.70.050.C, Shielding Standards, or be turned off no later than 30 minutes after the business closes.

K. **Internally Illuminated Architectural Elements.** Any architectural element, including walls or portions of buildings, including canopy facias, that is internally illuminated and that is not a sign or fenestration (windows or doors), shall have 100 percent of the initial lamp output of all lamps or Luminous Tubes used to provide such illumination counted toward partially shielded lighting for the purposes of calculating Total Outdoor Light Output for the site and is subject to the standards of subsection B. of this section.

L. **Architectural/Landscape Lighting.** Architectural lighting used to illuminate a structure or landscape lighting used to illuminate trees or other landscape elements is allowed subject to the following:

1. Architectural and landscape lighting that is directed downward onto a structure, tree, or other landscape feature shall be included in the Total Outdoor Light Output standards provided in Table 10-50.70.050.B, Maximum Total Outdoor Light Output Standards, based on whether a Fully Shielded or Partially Shielded Light Fixture is used; and
2. Architectural and landscape lighting that is directed upward onto a structure, tree, or other landscape feature is not allowed.

M. **Emergency Lighting.** Emergency lighting that is only turned on in the event of a power failure or when an alarm is activated is allowed in all lighting zones and is excluded from the Total Outdoor Light Output standards provided in Table 10-50.70.050.B, Maximum Total Outdoor Light Output Standards.

N. **Use of Mercury Vapor Outdoor Light Fixtures.** Except for outdoor lighting systems erected prior to 1950, no mercury vapor Outdoor Light Fixtures are allowed within the City of Flagstaff.

(Ord. 2026-01, Amended, 2/3/2026 (Res. 2026-03); Ord. 2022-27, Amended, 11/15/2022 (Res. 2022-51); Ord. 2016-07, Amended, 2/16/2016 (Res. 2016-02))

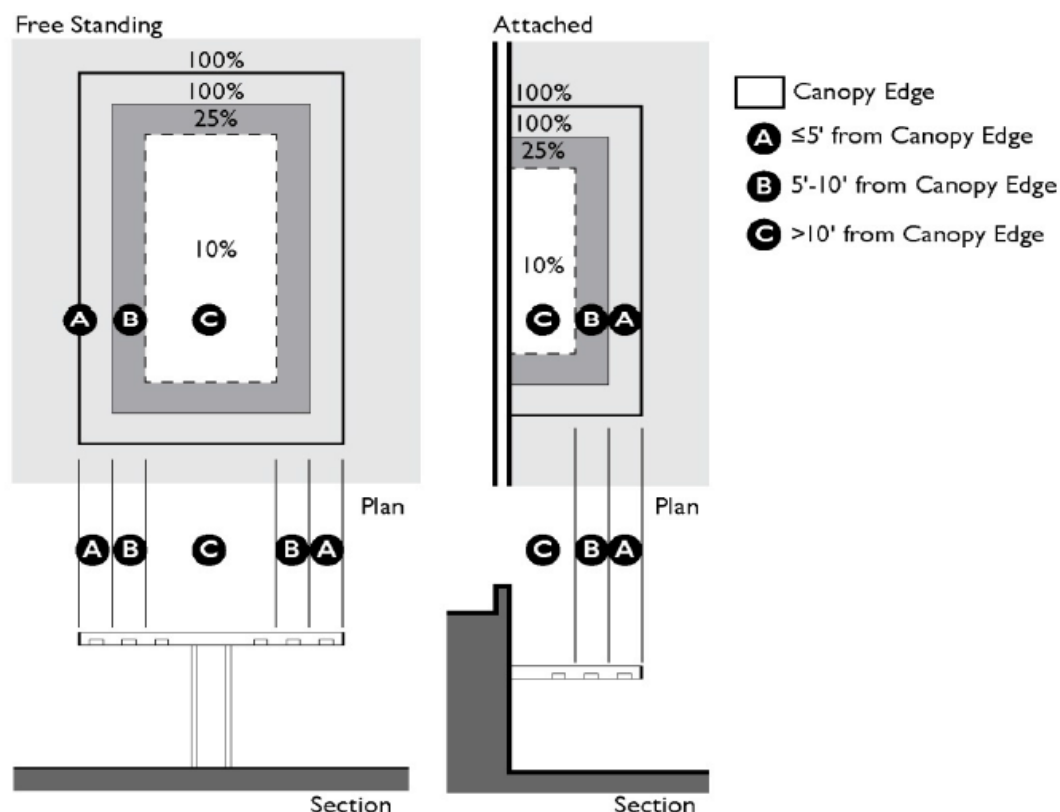
10-50.70.060 Special Uses

A. **Canopies, Building Overhangs, and Roof Eave Lighting.**

1. **Shielding.** All Outdoor Light Fixtures attached, mounted to, or within a canopy, building overhang, or roof eave shall be fully shielded.
2. **Total Lumens.** The Total Lumens of each Outdoor Light Fixture shall be calculated based on the distance from the edge of the canopy, building overhang, or roof eave subject to Figure 10-50.70.060.A.

Figure 10-50.70.060.A

Plan and Section Views of a Freestanding and Attached Canopy or Overhang, Showing Fixture Location and Initial Lamp Output Percentage Counted Toward Total Outdoor Light Output



B. Service Station Canopy Lighting.

1. **Lighting Class.** Lighting for service station canopies shall be considered Class 1 Lighting.
2. **Shielding.** All Outdoor Light Fixtures mounted on or recessed into the lower surface of service station canopies shall be fully shielded and utilize flat lenses. Such shielding must be provided by the fixture itself; shielding by surrounding structures such as canopy edges is not allowed.
3. **Total Under-Canopy Output.** The Total Outdoor Light Output used under service station canopies, defined as the sum of all under-canopy Light Fixtures, shall not exceed 30 Lumens per square foot of canopy in Lighting Zone 2, and shall not exceed 15 Lumens per square foot in Lighting Zone 1 (note: these values are not foot-candle illuminances).
4. **Total Lumens.** All lighting mounted under the canopy, including but not limited to Outdoor Light Fixtures mounted on or recessed into the lower surface of the canopy and any lighting within signage (but not including any lamps mounted within the pumps and used to illuminate information indicating such items as the total cost of fuel pumped and price per gallon), shall be included in the Total Outdoor Light Output for the site and is subject Figure 10-50.70.060.A.

C. Outdoor Recreation Facilities.

1. **Lighting Class.** Lighting for field/track/arena areas only shall be considered Class 1 Lighting.
2. **Lumen Cap Requirements.**
 - a. In Lighting Zone 1, lighting for field/track/arena areas is subject to the maximum Total Outdoor

Light Output standards set in Table 10-50.70.050.B;

b. In Lighting Zone 2, lighting for field/track/arena areas is not subject to the maximum Total Outdoor Light per acre limit set in Table 10-50.70.050.B; and

c. Illumination levels for the field/track/arena shall be designed to be no higher than recommended for Class IV play, as defined by the Illuminating Engineering Society of North America publication ANSI/IES RP-6-20, as amended.

3. **Shielding.** Fixtures used for field/track/arena areas shall be fully shielded.

4. **Time Limits.** Outdoor sports facilities shall not be illuminated after the time limits listed in Table 10-50.70.060.C, except to conclude a scheduled recreational or sporting event in progress prior to the time limitation.

Table 10-50.70.060.C		
Time Limits for Outdoor Lighting of Sports Facilities		
	Zone 1	Zone 2
Time limit when outdoor lights shall be turned off	9:00 p.m.	11:00 p.m.

5. **Certification.** Lighting systems for outdoor recreational facilities shall be designed and certified by an engineer registered in Arizona as conforming to all applicable restrictions of this code before construction commences. Further, after installation is complete, the system shall be again certified by a registered engineer to verify that the installation is consistent with the certified design.

D. **Street Lighting.**

1. Street lighting installed on public rights-of-way shall be in accordance with Title 13, Engineering Design Standards and Specifications for New Infrastructure.

2. Street lighting installed on private street tracts or easements shall be in accordance with Title 13, Engineering Design Standards and Specifications for New Infrastructure.

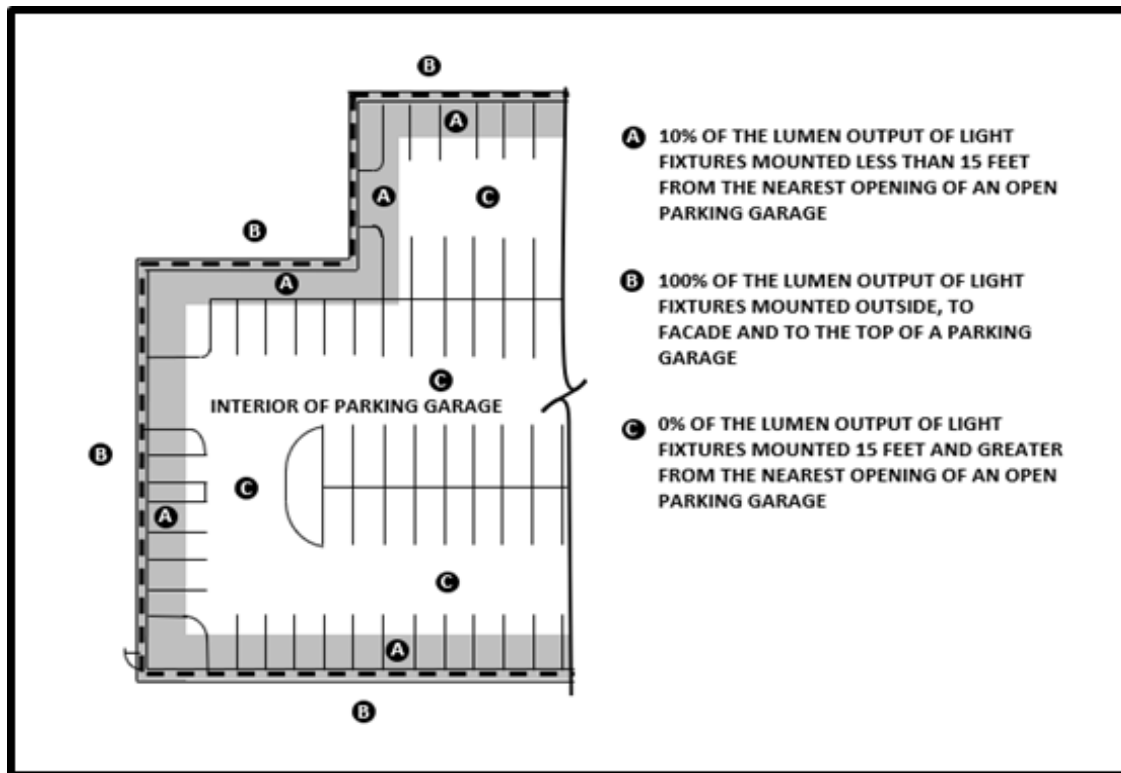
E. **Parking Garages.**

1. **Lighting Class.** Lighting installed for general illumination of parking areas within parking garages, where the parking areas are open to the outside, shall be considered Class 2.

2. **Total Outdoor Light Output.** Ten percent of the Lumen output of Light Fixtures mounted less than 15 feet from the nearest opening to the outdoors of an open parking garage shall be included toward the Total Outdoor Light Output standards established in Table 10-50.70.050.B subject to Figure 10-50.70.060.E. The Lumen output of Light Fixtures mounted 15 feet or more from the nearest opening to the outdoors of an open parking garage shall not be included toward the Total Outdoor Light Output standards in Table 10-50.70.050.B subject to Figure 10-50.70.060.E.

Figure 10-50.70.060.E

Calculation of Total Lumens of the Light Fixtures Located Inside of Parking Garage



3. **Shielding.** All light fixtures used on or within open parking garages, including those mounted to the ceilings over the parking decks, shall be fully shielded.

F. Outdoor Display Lots.

1. **Lighting Class.** Class 1 Lighting may be used to illuminate Outdoor Display Lots, and shall be included toward the Total Outdoor Light Output standards in Table 10-50.70.050.B.
2. **Lighting Time Limitations.** All Class 1 outdoor display lighting shall be turned off no later than 30 minutes after the close of business. Only lighting meeting Class 2 Lighting standards may be used to illuminate an outdoor display area 30 minutes or later after the close of business.
3. **Shielding.** All light fixtures used in Outdoor Display lots shall be fully shielded and be aimed so that the direct illumination shall be confined to the property boundaries of the source.

G. **Temporary Lighting.** Temporary lighting which does not conform to the provisions of this division may be approved by the Director subject to the approval of an outdoor lighting permit in accordance with the provisions of Section [10-20.40.100](#), Outdoor Lighting Permits. Temporary lighting is intended for uses which by their nature are of limited duration, including for example, holiday decorations, civic events, or construction projects.

(Ord. 2022-27, Amended, 11/15/2022 (Res. 2022-51); Ord. 2016-07, Amended, 2/16/2016 (Res. 2016-02))

10-50.70.070 Prohibited Outdoor Lighting

The following types of outdoor lighting are prohibited:

- A. Outdoor floodlighting by flood light projection above the horizontal plane.

B. Search lights, flood lights, laser source lights, or any similar high intensity light, except in emergencies by police, fire or medical personnel or at their direction; or for meteorological data gathering purposes.

C. Any lighting device located on the exterior of a building or on the inside of a window which is visible beyond the boundaries of the lot or parcel with intermittent fading, flashing, blinking, rotating or strobe light illumination.

(Ord. 2022-27, Amended, 11/15/2022 (Res. 2022-51))

10-50.70.080 Exceptions

A. **Flagstaff Municipal Airport Lighting.** Required navigational lighting systems at public airports for the safe and efficient movement of aircraft during flight, takeoff, landing, and taxiing are exempt from the provisions of this division. Lighting used for illumination of aircraft loading, unloading, and servicing areas is exempt from the Lumens per acre limits provided in Table 10-50.70.050.B., although it must conform to all other requirements of this division. All other outdoor lighting at public airport facilities shall comply with the provisions of this division. This provision does not apply to private helipads or landing strips.

B. **Infrared Security Lighting.** Lights emitting infrared radiation used for remote security surveillance systems are permitted in all zones with the following restrictions:

1. Fixed lights must be fully shielded; and
2. Movable lights, such as spot lights attached to infrared sensitive cameras, must be mounted such that the lights cannot be directed higher than 20 degrees below the horizontal, measured from the center of the light beam.

C. **Emergency Lighting by Emergency Services.** Searchlights, floodlights, laser source lights, strobe or flashing lights, or any similar high intensity lights are permitted when used in emergencies by police, fire, medical, or utility personnel or at their direction.

D. **Holiday Decorations.** In all lighting zones, low-Lumen output holiday decorations may be unshielded and remain on all night from November 15th to January 15th.

E. **Solar-Powered Lighting.** Solar-powered lights of five watts or less per fixture used in residential landscaping applications and to illuminate walkways are exempt from applicable lamp type and shielding standards and are excluded from the total Lumen calculations for the site.

F. **Construction and Renovation of Municipal Facilities.** All outdoor lighting used for construction or major renovation of municipal buildings, structures and facilities is exempt from the provisions of this division.

(Ord. 2022-27, Amended, 11/15/2022 (Res. 2022-51))

10-50.70.090 Nonconforming Outdoor Lighting

Section [10-20.60.080](#), Nonconforming Outdoor Lighting, provides the standards and regulations for nonconforming outdoor lighting.

(Ord. 2022-27, Amended, 11/15/2022 (Res. 2022-51))

10-50.70.100 Violations and Enforcement

A. It shall be unlawful to install or operate an Outdoor Light Fixture in violation of this division. Any person violating any provisions of this division shall be guilty of a misdemeanor. Each and every day during which the illegal erection, maintenance, and use continues shall be considered a separate offense.

B. The requirements of this division shall be enforced in compliance with the enforcement provisions of Division [10-20.110](#), Enforcement.

(Ord. 2022-27, Amended, 11/15/2022 (Res. 2022-51))

The Flagstaff City Charter and City Code are current through Ordinance 2026-01, passed February 3, 2026.

Disclaimer: The City Clerk's Office has the official version of the Flagstaff City Code. Users should contact the City Clerk's Office for ordinances passed subsequent to the ordinance cited above.
